Online data sheet

S710 E 1217512

S700

EXTRACTIVE GAS ANALYZERS

SICK
Sensor Intelligence.

Ordering information

Type	Part no.
S710 E 1217512	1217512

Other models and accessories ➜ www.sick.com/S700

Detailed technical data

Technical specifications

Description	19″ rack enclosure with 3 rack units, for integration in cabinets
Measured values	CO, NO, O_2, SO_2
Performance-tested measurands	CO, NO, O_2, SO_2
Maximum number of measurands	4
Measurement principles	NDIR spectroscopy, electrochemical cell
Gas flow rate	
	5 l/h ... 100 l/h
Required operating range:	30 l/h ... 60 l/h
Measuring ranges	
CO	0 ... 200 mg/m³ / 0 ... 2,000 mg/m³
NO	0 ... 250 mg/m³ / 0 ... 2,500 mg/m³
O_2	0 ... 10 Vol.-% / 0 ... 25 Vol.-%
SO_2	0 ... 250 mg/m³ / 0 ... 2,500 mg/m³
Certified measuring ranges	
CO	0 ... 200 mg/m³
NO	0 ... 250 mg/m³
O_2	0 ... 25 Vol.-%
SO_2	0 ... 250 mg/m³
Response time (t_{90})	MULTOR analyzer module: ≤ 25 s At 60 l/h, depending on cuvette length, gas flow and number of measuring components OXOR-E analyzer module: 20 s Typical at 60 l/h, depending on gas flow
Sensitivity drift	
MULTOR analyzer module	≤ 1 %: of measuring range full scale per week
OXOR-E analyzer module	≤ 1 %: of measuring range full scale per week
Zero point drift	
MULTOR analyzer module	≤ 1 %: of smallest measuring range per week
OXOR-E analyzer module	≤ 2 %: of smallest measuring range per month
Sample gas temperature	

Analyzer inlet:	0 °C ... +45 °C
Process pressure	
Hosed gas lines:	–200 hPa ... 300 hPa
Process gas humidity	Non-condensing
Dust load	Free of dust and aerosols
Ambient temperature	+5 °C ... +45 °C
Storage temperature	–20 °C ... +70 °C
Ambient pressure	700 hPa ... 1,200 hPa
Geographical altitude	≤ 2,000 m (above mean sea level)
Ambient humidity	≤ 95 % Relative humidity; non-condensing
Conformities	Approved for plants requiring approval 2001/80/EC (13. BImSchV) 2000/76/EC (17. BImSchV) 27.BImSchV TA-Luft (Prevention of Air Pollution) EN 14181
Electrical safety	CE
Enclosure rating	IP20
Analog outputs	4 outputs: 0/4 ... 20 mA, 500 Ω
Digital outputs	8 relay contacts: Three relay outputs preset for failure, service and maintenance 8 Open collector outputs: Freely adjustable
Digital inputs	8 optical coupler inputs: Electrically isolated; freely programmable
Modbus	✓
Type of fieldbus integration	RTU RS-232
Indication	LC display
Operation	Menu-driven operation via LC-display and membrane keyboard
Menu languages	German, English, French, Italian, Dutch, Polish, Swedish, Spanish
Dimensions (W x H x D)	483 mm x 132.5 mm x 390 mm For details see dimensional drawings 483 mm x 132.5 mm x 390 mm (for details see dimensional drawings)
Weight	10 kg ... 20 kg Depending on configuration

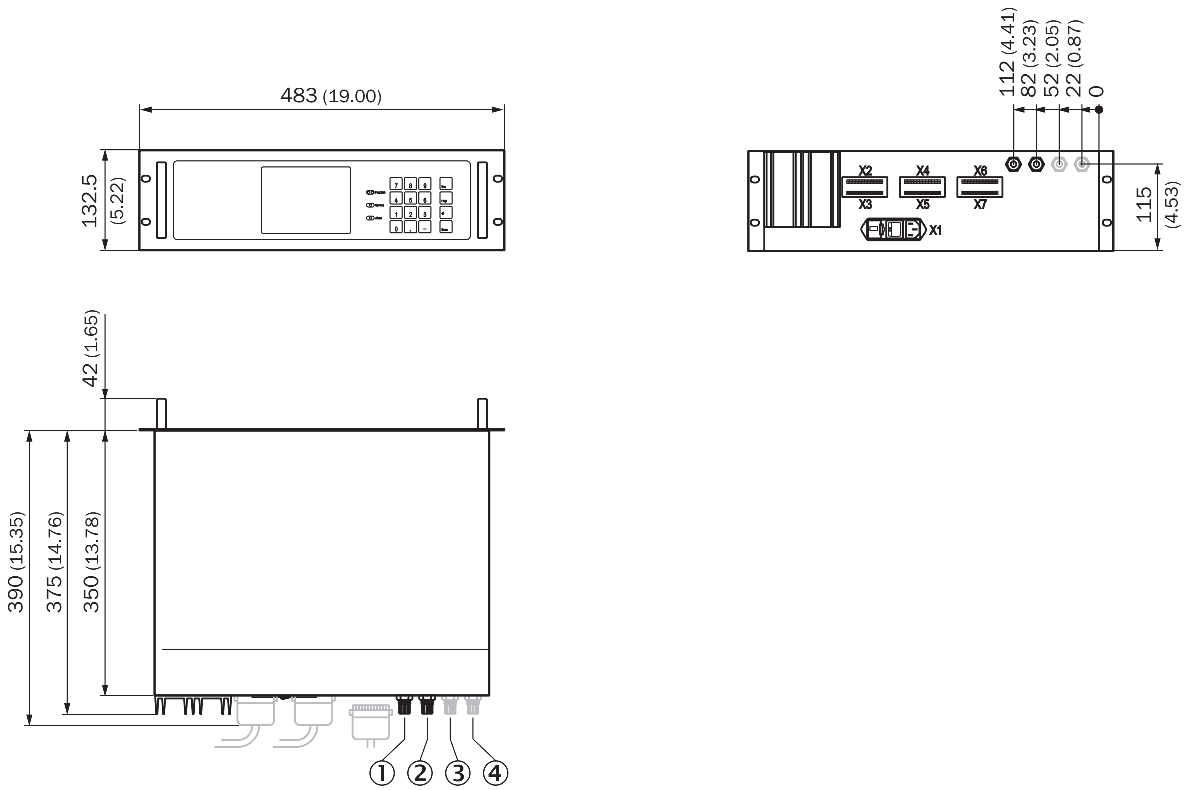
Material in contact with media	Viton B, PVDF, CaF_2, gold, Aluminum
Power supply	
Voltage	100 V AC / 115 V AC / 230 V AC
Frequency	48 ... 62 Hz
Power consumption	≤ 150 W Depending on system configuration
Sample gas connections	 PVDF bulkhead fitting
Corrective functions	Automatic testing and adjustment with test gases Manual adjustment with test gases
Integrated components	MULTOR analyzer module OXOR-E analyzer module PVDF bulkhead fitting

Classifications

ECl@ss 5.0	27150302
ECl@ss 5.1.4	27150302
ECl@ss 6.0	27150302
ECl@ss 6.2	27150302
ECl@ss 7.0	27150302
ECl@ss 8.0	27150302
ECl@ss 8.1	27150302
ECl@ss 9.0	27150302
ETIM 5.0	EC001190
ETIM 6.0	EC001190
UNSPSC 16.0901	41115406

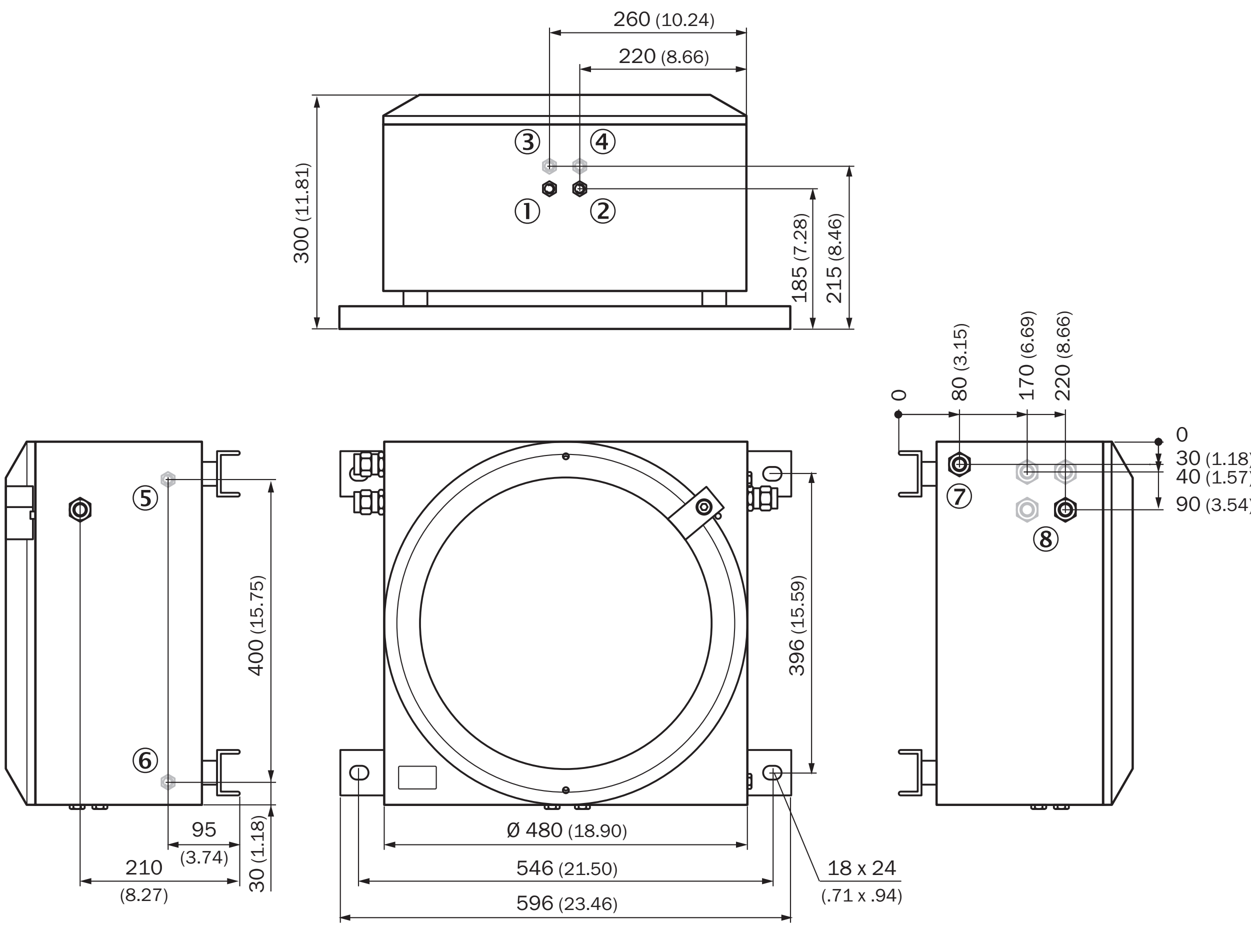
Dimensional drawing (Dimensions in mm (inch))

S710 design



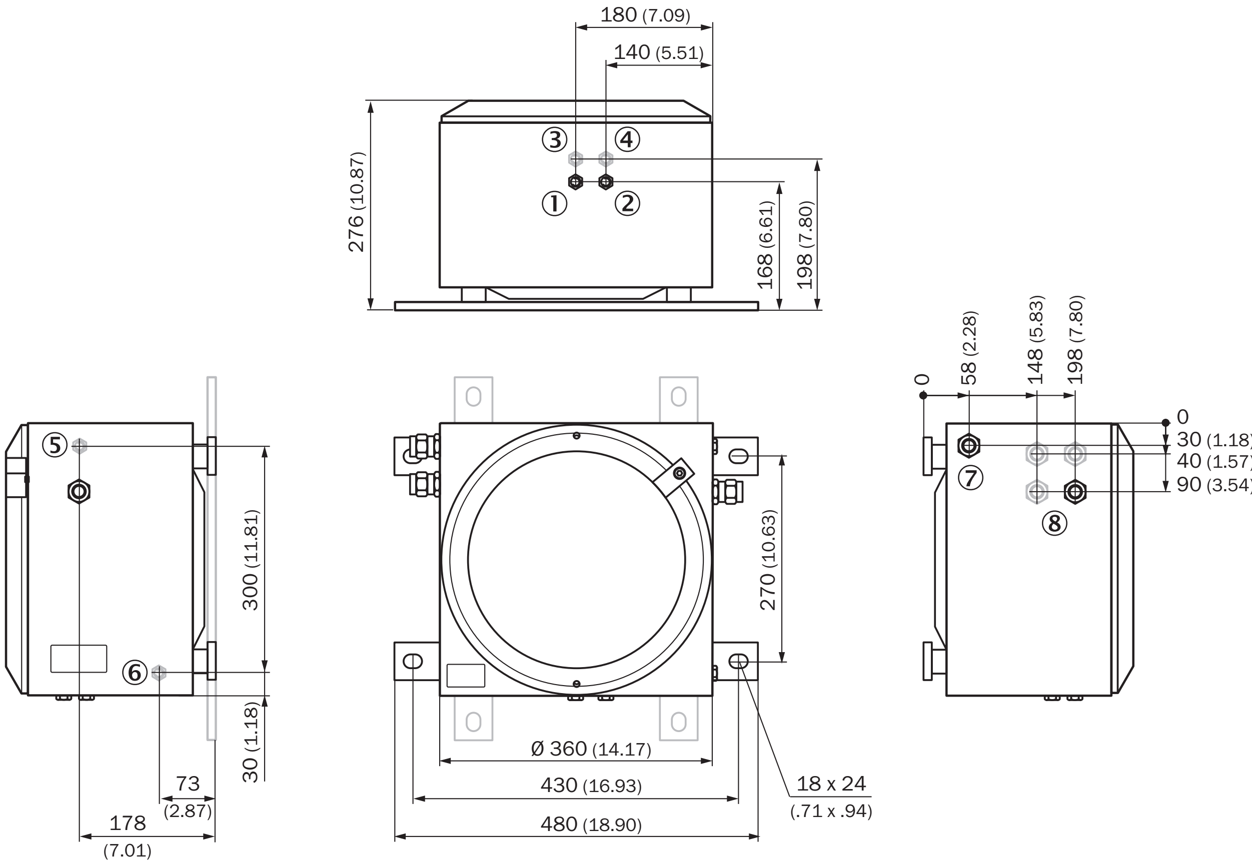
① 1. sample gas inlet
② Exhaust gas outlet
③ 2. sample gas inlet
④ 3. sample gas inlet

S721 Ex design



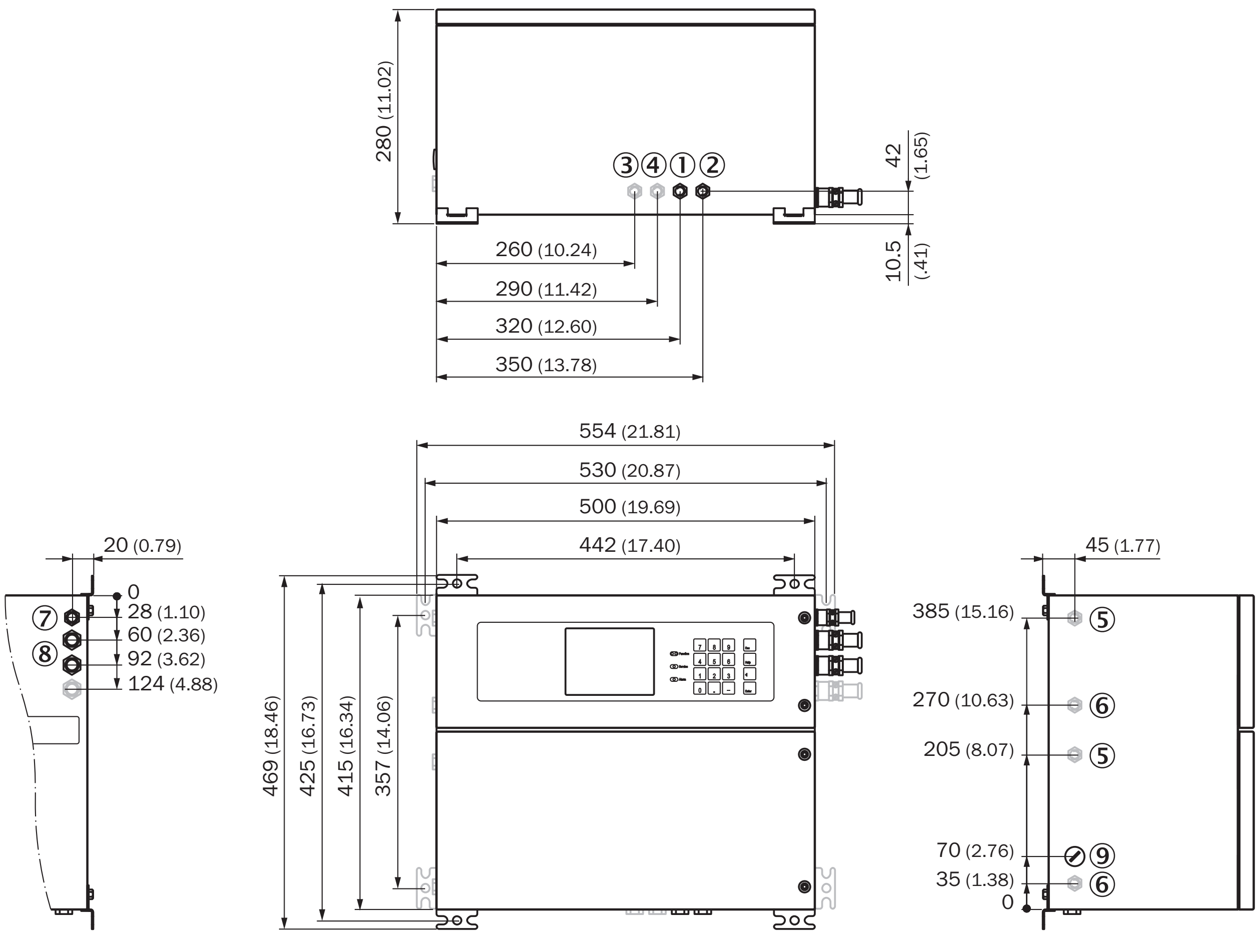
① 1. sample gas inlet
② Exhaust gas outlet
③ 2. sample gas inlet
④ 3. sample gas inlet
⑤ Purge gas inlet
⑥ Purge gas outlet

S720 Ex design



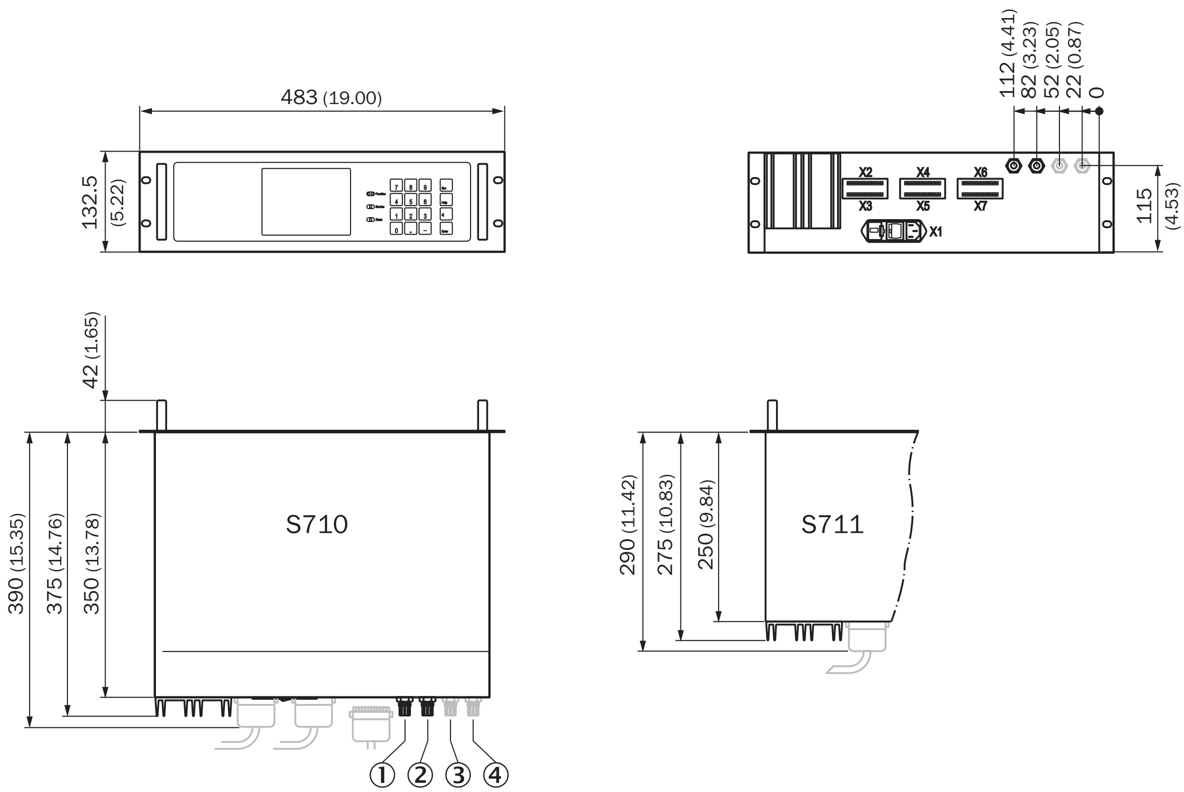
① 1. sample gas inlet
② Exhaust gas outlet
③ 2. sample gas inlet
④ 3. sample gas inlet
⑤ Purge gas inlet
⑥ Purge gas outlet

S715 design



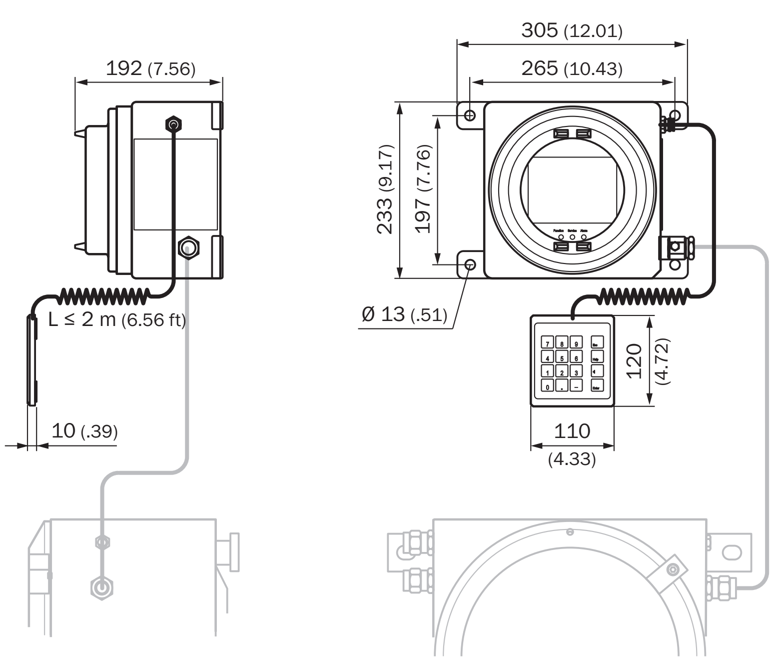
① 1. sample gas inlet
② Exhaust gas outlet
③ 2. sample gas inlet
④ 3. sample gas inlet
⑤ Purge gas inlet
⑥ Purge gas outlet

S710 and S711 design



① 1. sample gas inlet
② Exhaust gas outlet
③ 2. sample gas inlet
④ 3. sample gas inlet

Display and keyboard for S720 Ex and S721 Ex design



SICK AT A GLANCE

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We have extensive experience in a wide range of industries and understand their processes and requirements. With intelligent sensors, we can deliver exactly what our customers need. In application centers in Europe, Asia and North America, system solutions are tested and optimized in accordance with customer specifications. All this makes us a reliable supplier and development partner.

Comprehensive services complete our offering: SICK LifeTime Services provide support throughout the machine life cycle and ensure safety and productivity.

For us, that is "Sensor Intelligence."

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